A Study of Active Galactic Nuclei With SDSS DR7

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Abstract

It has long been established that massive galaxies hosting accreting supermassive black holes produce extreme emission lines from their centers. These centers are incredibly small (usually only a bit larger than the solar system), yet produce enough energy to match or outshine the luminosity of the galaxy as a whole. This study will examine the emission line properties of galaxies and determine if they are hosting these active galactic nuclei (AGN), and compare certain properties to those of normal star-forming galaxies. It can be determined through analysis of emission line ratios that AGN make up 14.2% of galaxies in the local universe, and through histogram visualization, that they have higher mean stellar mass and higher mean [OIII] luminosity.

1 An Introduction

1.1 What are Active Galactic Nuclei?

1908 saw the very first discovery of a very peculiar galaxy, as astronomer Edward A. Fath was conducting long exposure photographic spectra studies of possible extragalactic objects[1]. He wanted to test whether these objects, now known to be spiral galaxies, would produce stellar absorption lines consistent with a collection of stars, and for the large majority, this was true. But for one particular object, NGC 1068, it showed not only absorption lines but very bright emission lines.

NGC 1068 would later prove to be the first documented case of an active galactic nucleus (AGN), some of the most persistently energetic sources in the universe. The term currently refers to a compact central region of a galaxy that produces extremely luminous and high energy phenomena caused by an accreting supermassive black hole (SMBH). The SMBH is an engine that fuels intense energy production, and results in features such as the relativistic jets, winds, and broad UV/optical emission lines seen in the host galaxies. They have become an area of intense study for astrophysicists, as they provide a unique environment for high energy physics and galactic evolution.

How, though, are we able to tell which galaxies truly host these strong central engines? Distinguishing AGN from regular star-forming galaxies is, even still, a heavily debated and studied topic. Differing detection methods lead to a patchwork of confusing classification schemes, which can obscure the true mechanism linking these excited galaxies together. Star-forming galaxies then get mixed up easily with AGN, through factors such as extreme photoionization of surrounding gas

[1] At the time, spiral galaxies were not yet proved to be their own "island universes," they were still widely believed to be nebulae within the Milky Way. Fath was one of the first to note the true nature of these spiral nebulae.

from O and B stars in extreme starburst galaxies, shock-wave heating, or photoionization from the power-law continuum (Baldwin et al. 1981).

That is to say, it becomes extremely important to find a more concrete way of classifying active galaxies. Baldwin et al. (1981) proposed a classification scheme for the strength of excitation in a galaxy based on emission line ratios, with Kauffmann et al. (2003) and Kewley et al. (2001) later proposing demarcations to separate galaxies into distinct classes. This study will apply these schemes to a sample of galaxies, and examine their properties in comparison to known characteristics of AGN, in order to determine whether we truly can distinguish these massive, high-energy nuclei.

1.2 Areas of Study

This study will mainly focus on the identification of AGN through emission line ratios. There will also be an examination of the properties of the identified AGN and a comparison of them with proper star-forming galaxies.

Starburst galaxies produce extremely strong emission lines of highly ionized species, such as [Fe VIII] or [Ne V], which need a huge amount of energy to lose so many electrons. One of the strongest indicators specifically that a galaxy is hosting an AGN, rather than just being a normal starburst galaxy, is the strength of its Balmer lines. Any starburst galaxy or AGN will intrinsically have a Hα to Hβ line ratio of 3, but AGN will have much stronger and sometimes broader lines than that of a starburst. However, because Hβ struggles to escape from dust-heavy regions, the extinction of Hβ is much higher than that of Hα, leading to a much larger observed Hα / Hβ ratio, with the Hα line sometimes being greater than 5 times stronger than the Hβ line.

In order to counteract the effects of dust, "extinction independent" line ratios are chosen to discriminate against the excitation mechanism of the galaxy. These species experience extinction extremely similarly to Hα and Hβ. The standard emission lines ratios chosen are [OIII]/Hβ, [NII]/Hα, [SII]/α, and [OI]/Hα, with the brackets denoting forbidden lines. This study will focus on the line ratios [OIII]/Hβ and [NII]/Hα, following Kauffman et al. (2003).

There is certain criteria that the emission line ratios of a galaxy must meet in order to be classified as hosting an AGN, the most widely established being that of Kauffman et al. (2003) and Kewley et al. (2001). In order to visualize the data classification, BPT diagrams will be made following the standards in these papers. BPT diagrams refer to the diagrams laid out by Baldwin, Phillips, & Terlevich (1981) that have the emission line ratios on each axis and allow for the easy catagorization of galaxies depending on where they lie on the diagram. The specific classification scheme used in this study does not follow the Baldwin, Phillips, & Terlevich paper.

After classification, the two chosen galactic properties that will be considered and compared are the stellar masses and luminosity of the [OIII] line, which can track the strength of activity in the nucleus.

The forbidden [OIII] line is specifically chosen because it is a good tracer of strength of activity in the nucleus. According to the standard unified AGN model discussed in the Kauffman paper, AGN can be classified entirely depending on the viewing orientation, as seen in Figure (1). The dusty torus obscuring the nucleus allows a small amount of radiation to escape and photoionize the surrounding gas clouds, leading to strong permitted and forbidden lines in the narrow-line region such as [OIII]. It therefore allows an isotropic view of the central engine from outside of the main region of activity, and is much easier to detect than possible other lines. The implicit argument here is the the [OIII] lines are then a very good tracer for the strength within that main region, and they are AGN ionized, rather than star-formation ionized, above the demarcation line.

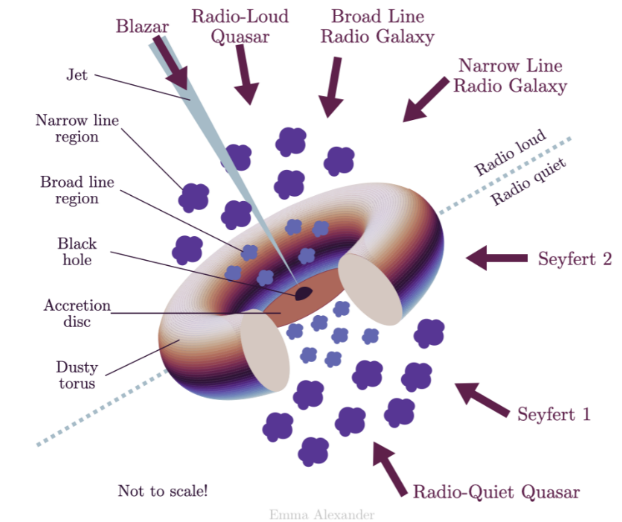


Figure 1: AGN Unified Model

2 The Spectroscopic & Photometric Sample

This study needs a large and uniform sample of galaxies with both spectroscopic and photometric abilities, which is why the Sloane Digital Sky Survey (SDSS) public data access catalogs were chosen. SDSS provides very high quality photometric and spectroscopic data of a very large sample of galaxies. In particular, Data Release 7 (DR7), which was in collaboration with MPA/JHU, had a focus on the statistical properties of galaxies and AGN, with value-added catalogs specifically designed for these studies. It has large datasets that cover over 45,000 square degrees of sky and over 1 million galaxies, containing in-depth absorption and emission line spectra, stellar masses, and star formation rates.

2.1 SDSS Telescope

The survey utilized a modified Ritchey-Chretien telescope at the Apache Point Observatory with imaging and spectroscopy abilities – a large-mosaic CCD camera that can image with 5 optical bands and two spectrographs (one with a blue channel and one with a red channel). The CCDs are specifically set up to in six columns of five each, with each row corresponding to the SDSS r, i , u, z, and g filters. The telescope used a drift scan mode that has the telescope moving in circles across the sky that has images of objects being collected at the same rate as the CCD data is being read. This allows for five images of each object (one per filter). The spectroscopic instrument has cartridges containing plates with drilled holes plugged with optical fibers mounted onto the telescope. Exposures are collected on the plates, at least three per object.

2.2 Data Processing

SDSS then did raw data processing, and released the raw data in FITS files, which MPA/JHU used to create derived datasets. The derived datasets contain gas-phase metallicity estimates, star formation rates, and stellar masses. Stellar mass calculations follow Kaufmann et al (2003), star formation rates calculations follow Brinchmann et al (2004), and gas phase metallicities follow Tremonti et al (2004).

gal_info	gal_line	stellarmasses
RA	OIII_5007_FLUX	MEDIAN
DEC	OIII_5007_FLUX_ERR	P16[1]
Z	H_BETA_FLUX	P84[2]
Z_WARNING	H_BETA_FLUX_ERR	
SN_MEDIAN	NII_6584_FLUX	
	NII_6584_FLUX_ERR	
	H_ALPHA_FLUX	
	H_ALPHA_FLUX_ERR	

Figure 2: Chosen Columns from `gal_info`, `gal_line`, and `stellarmasses` Files

2.3 The Catalogs

The files used are the `gal_info`, `gal_line`, and `stellarmasses` files as found on the MPA/JHU DR7 Release of Spectrum Measurements website. This website contains all the value added galaxy catalogs as custom galaxy spectral fitting codes of raw data that includes line fluxes, line widths, etc. Both `gal_info` and `gal_line` are part of the raw data FITS files. `Stellarmasses` is a derived data FITS file.

`Gal_info` contains SDSS targeted info from the plugmap and redshift and velocity dispersion, and gives redshifts, velocity dispersions, right ascension, declination. It was chosen due to its redshift columns, which are needed for later luminosity calculations. `Gal_line` contains width and flux measurements of emission lines for 20 different lines in CGS units. `Stellarmasses`, the only file from the derived data, contains log masses calculated from *photometry*, as well as percentile measurements. These files are all integral to calculations that will be done later in this study.

All data analysis was conducted in Python, using astropy, numpy, and matplotlib libraries. In order to sufficiently analyze the data, specific columns were picked from each file and combined into one singular dataframe using astropy.table. Figure (2) depicts which columns were chosen from each file to analyze.

Emission line columns were chosen according to Veilleux & Osterbrock (1986) line ratios of [OIII] λ5007 / Hβ and [NII] λ6583 / Hα. Veilleux & Osterbrock follow five criteria for best line ratios for AGN classification, including:

1. Ratios must be made up of strong lines easy to measure in typical spectra.
2. Lines cannot be badly blended with other lines as deblending increases uncertainty.
3. Wavelength separation must be small as to avoid reddening and flux discrepancies.
4. Ratios should contain one element and one HI Balmer line as these are less likely to depend on chemical composition of the galaxy.
5. Lines need to be easily read with present day instrumentation.[2]

[2] "Present day instrumentation" here is referring to instrumentation of the year 1986, when Veilleux & Osterbrock was published. By the SDSS DR7 release in 2008, instrumentation had gotten increasingly more sensitive. Nonetheless, the emission lines still follow the criterion as written by Veilleux & Osterbrock to ensure higher accuracy.

While there are multiple different possible line ratios that fall into these criterion, [OIII] λ5007 / Hβ and [NII] λ6583 / Hα are the strongest candidates, and also follow both Kauffman et al. (2003) and Kewley et al. (2001). These lines specifically are almost completely insensitive to reddening and/or spectrophotometric errors.

The data was then quality cut in order to maximize accuracy. Following Kauffman et al., a signal to noise ratio (S/N) was chosen to be S/N > 3 for all four emission lines by $S/N = \frac{flux}{err} > 3$. Redshifts were filtered by the `Z_WARNING` column with bad redshifts having `Z_WARNING` $\neq 0$, which was calculated by MPA/JHU using the Schlegel data model.

Quality cuts also included a chosen redshift range. The redshift range is very important for calculations using emission line fluxes. Higher redshift galaxies have more redshifted emission lines, creating discrepancies in wavelength and which filters can capture the spectrum. Luckily, these catalogs had this issue accounted for by the SDSS data processing team for emission line fluxes and errors, so this would not affect any data analysis in this study. However, a larger issue is that galaxies in different redshift bins are captured in different galactic eras. Much higher redshifted galaxies are still in the early ages of galaxy formation, and therefore AGN would have differing properties. As AGN properties in different redshift bins is not something that is going to analyzed in this study, a low redshift range of 0.04 < z < 0.2 was chosen, which follows Kauffman et al.

After quality cuts, 320,759 galaxies were left of the original 927,552 (34.6%).

3 Data Analysis

3.1 Classifications

Classifications were based on Kewley et al. (2001) and Kauffman et al. (2003). Kewley et al. defines a theoretical upper limit for a starburst model in which anything above or to the right of the empirical limit is not a starburst by any current models of metallicity and ionization parameters. Galaxies above this line cannot be explained by any possible combination of parameters, and no normal star-forming HII region can exist above this line, so it must be explained by something else. The limit is defined as

$$\log_{10}([OIII]/H\beta) = \frac{0.61}{\log_{10}([NII]/H\alpha) - 0.47} + 1.19 \tag{1}$$

Kauffman et al., on the other hand, defines a much lower demarcation line between starburst galaxies and AGN. They used very similar, if less conservative, reasoning as Kewley et al., and had an empirical limit defined as

$$\log_{10}([OIII]/H\beta) = \frac{0.61}{\log_{10}([NII]/H\alpha) - 0.05} + 1.3 \tag{2}$$

with galaxies defined to be possible AGN if they satisfy

$$\log_{10}([OIII]/H\beta) > \frac{0.61}{\log_{10}([NII]/H\alpha) - 0.05} + 1.3 \tag{3}$$

For the classification in this study, the Kewley limit was used as an upper limit on star-forming galaxies while the Kauffman limit was used as the lower limit for AGN. Galaxies were defined to be star-forming if they had

$$\log_{10}([OIII]/H\beta) < \frac{0.61}{\log_{10}([NII]/H\alpha) - 0.05} + 1.3 \quad \text{(The Kauffmann Limit)} \tag{4}$$

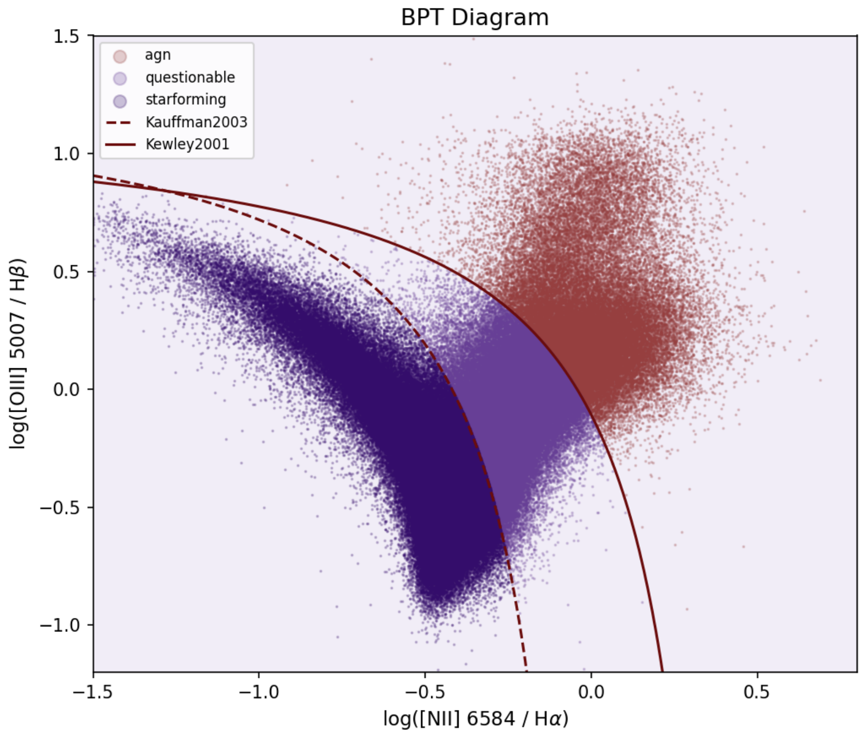


Figure 3: BPT Line Ratio Classification Diagram

and defined to be AGN if they had

$$\log_{10}([OIII]/H\beta) > \frac{0.61}{\log_{10}([NII]/H\alpha) - 0.47} + 1.19 \quad \text{(The Kewley Limit)} \tag{5}$$

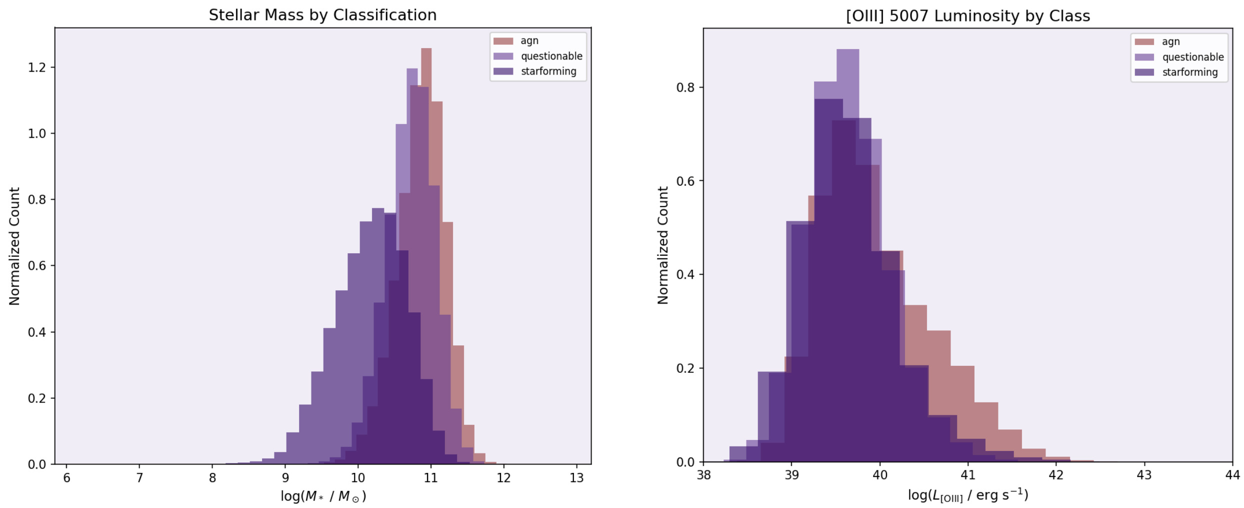
For galaxies in between the two lines, or galaxies that were neither defined to be star-forming nor AGN, they were classified as a composite group called "questionable". Questionable galaxies had spectroscopic properties that were classified differently depending on which model was used, and could be either high luminosity star-forming or lower luminosity AGN. Though this study will not probe the questionable group in much detail, a follow up in future work could be use other methods to distiguish the AGN from star-forming gaalxies within this group, using UV continuum, mid-IR excess, radio detections, or spectral line width.

Figure (3) depicts the resulting BPT diagram.

It is also worth noting that though the Kauffman and Kewley equations define certain spectroscopic properties to be the result of AGN only, they could also be the results of shocks. By Kaschinski & Ercolano (2013), shocks are unstable radiation-driven winds that are a result of EUV and X-ray radiation produced from cooling zones. They disrupt spectral energy distributions and produce extra emission lines (though it depends on temperature), resulting in differing photoionization models with shock stellar atmosphere models as an energy source. As a product of this, the AGN classification schema could be affected.

3.2 Properties and Comparisons

In order to study how galactic properties differ between the modeled classes, two particular characteristics were chosen to reveal distribution discrepancies between classified AGN and normal



(a) Stellar Mass Distribution by Class

(b) [OIII] Luminosity Distribution by Class

Figure 4: Galactic Property Comparison Histograms

star-forming galaxies: stellar mass distribution and [OIII] λ5007 emission line luminosity distribution.

The stellar mass `MEDIAN` column was made into an array of $\log_{10}$ masses, and then divided further into each galaxy's classification. Distributions were plotted in a histogram as seen in Figure (4a).

The [OIII] λ5007 emission line luminosity distribution needed distance calculations. Distances were derived from redshifts using astropy.cosmology and assuming a Flat Lambda CDM cosmological model (H0 = 70, Om0 = 0.3). Luminosity was then found using the log of the classic equation relating flux, distance, and luminosity, $Flux = \frac{L}{4\pi d^2}$, with L as luminosity and d as the derived distances. Before calculation, fluxes were converted to standard CGS units, as `gal_line` has all flux measurements in $1 \times 10^{-17} erg/s/cm^2$. The final calculated luminosities were then made into an array and further divided by each galaxy's classification. A histogram was made depicting the distribution, as seen in Figure (4b).

3.3 Final Measurements by Class

Classification counts and percent fractions of the total sample[3] are depicted in Figure (5). Stellar mass distributions by class, including the median, mean, and standard deviation, is depicted in Figure (6). [OIII] emission line luminosity distribution by class, including the median, mean, and standard deviation, is depicted in Figure (7).

CLASS	COUNT	FRACTION
Star Forming	202,076	63.0%
Questionable	73,050	22.8%
AGN	45,633	14.2%

Figure 5: Classification Counts and Fractions

[3]The quality cut sample of 320,759 galaxies.

CLASS	MEDIAN	MEAN	STD
Star Forming	10.20	10.17	0.50
Questionable	10.74	10.72	0.34
AGN	10.89	10.86	0.33

Figure 6: Stellar Mass Distributions

CLASS	MEDIAN	MEAN	STD
Star Forming	39.60	39.66	0.63
Questionable	39.62	39.65	0.47
AGN	39.84	39.96	0.64

Figure 7: [OIII] Luminosity Distributions

4 Discussion

The total AGN fraction of the quality cut section of 320,759 galaxies was 14.2%. It is still a heavily debated topic what the exact fraction is in the local universe, but the Kauffman paper defined 22,623 AGN out of 122,808 galaxies, which is $\approx 18\%$. The 4% discrepancy is likely because that study did not use the Kewley demarcation, which would greatly decrease the number of classified AGN. However, the actual BPT diagram is incredibly visually consistent with the Kauffman and Kewley BPT diagrams.

AGN were also more massive on average, peaking past $\log_{10}(\frac{M_*}{M_\odot}) = 11$, and had higher mean [OIII] luminosity, with a tail extending past $\log_{10} L_{[OIII]} = 42$.

Knowing that AGN are more massive than average star-forming galaxies is integral in understanding the physics behind AGN. Higher mass black holes are able to sustain higher accretion rates (and therefore produce much higher luminosities, or power output) before reaching their Eddington limit and blowing away all their fuel. Through the M - σ relation, central black hole mass directly scales to the host galaxy's mass. That is to say, in order for a galaxy's SMBH to match or even significantly outshine the rest of the galaxy (which would be an AGN), the galaxy must be of *very* high mass.

The [OIII] luminosity having a longer tail is also a very important conclusion. Recall that the [OIII] luminosity was a good tracer of strength in the nucleus because very high [OIII] emission comes from the ionization of gas clouds in the narrow-line region, which re-emits the radiation from the SMBH in the form of strong narrow emission lines. Star-forming galaxies are unable to emit [OIII] to this level of strength, and this emission luminosity in the AGN is almost always from the accreting SMBH.

In this regard, the conclusion can be made that the results found relatively match what is currently known about AGN. Therefore, the methods used in this study are strong techniques for categorization of galaxies.

Of course, the dataset was a very limited sample of galaxies because of the very small redshift range, and it would have been very interesting to look at a much larger sample and see the distributions of class, stellar mass, and [OIII] luminosity across redshift bins. Because this study only examined within the very local universe, it is limited in significant contributions to the overall understanding of cosmological evolution across time.

It is also important to consider that the “questionable” class was not examined. While these are very likely not AGN because of they do not produce strong enough Balmer line emissions, this composite group is still distinct from normal star-forming galaxies. Perhaps they are very massive starburst galaxies, or galaxies experiencing shocks, but an examination of this class and a deeper comparison to true AGN should be done in the future. Distinguishing these galaxies could be done through UV emission continuum analysis, mid-IR excess analysis, a comprehensive examination of the width of the spectral lines, or perhaps through radio detections.

However, the dataset chosen is one to the strongest for this sort of examination. It is thorough and the value-added catalogs provide very accurate data for all the needed probes. The Kauffman and Kewley equations are also very well-proven to be a good standard of AGN classification. These catalogs and equations have thoroughly increased the accuracy of this study.

Overall, the research done in this paper was a successful investigation into the fraction and properties of active galactic nuclei. These properties are very important for the understanding of black hole evolution and accretion, and how massive galaxies who host accreting supermassive black holes are affected. While the Milky Way itself is not massive enough to host an AGN, at only $\log_{10}(\frac{M_*}{M_\odot}) \cong 10.41$ according to Lian et al. (2025), it still hosts a SMBH, Sagittarius A*, at it's center. Having knowledge of the effects of extreme SMBH accretion in massive galaxies (AGN), allows us to approximate the effects on our own galaxy if Sagittarius A* begins to accrete huge amounts of mass. It is very crucial to emphasize the importance of AGN research in the future of astronomy and astrophysics as these objects truly hold so much information on galactic formation and evolution, and the cosmological web of our universe.

5 Images of Classified AGN Galaxies

Pictured below are images of 5 randomly selected AGN classified galaxies. These images are provided by the SDSS DR7 Navigation Tool. Right ascension and declination are included.

Figure 8: ra = 209.77393, dec = 55.699425

Figure 10: ra = 196.73401, dec = 55.94665

Figure 9: ra = 145.68114, dec = -0.86722344

Figure 11: ra = 212.0057, 14.986399

Figure 12: ra = 191.28258, dec = 57.134197

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